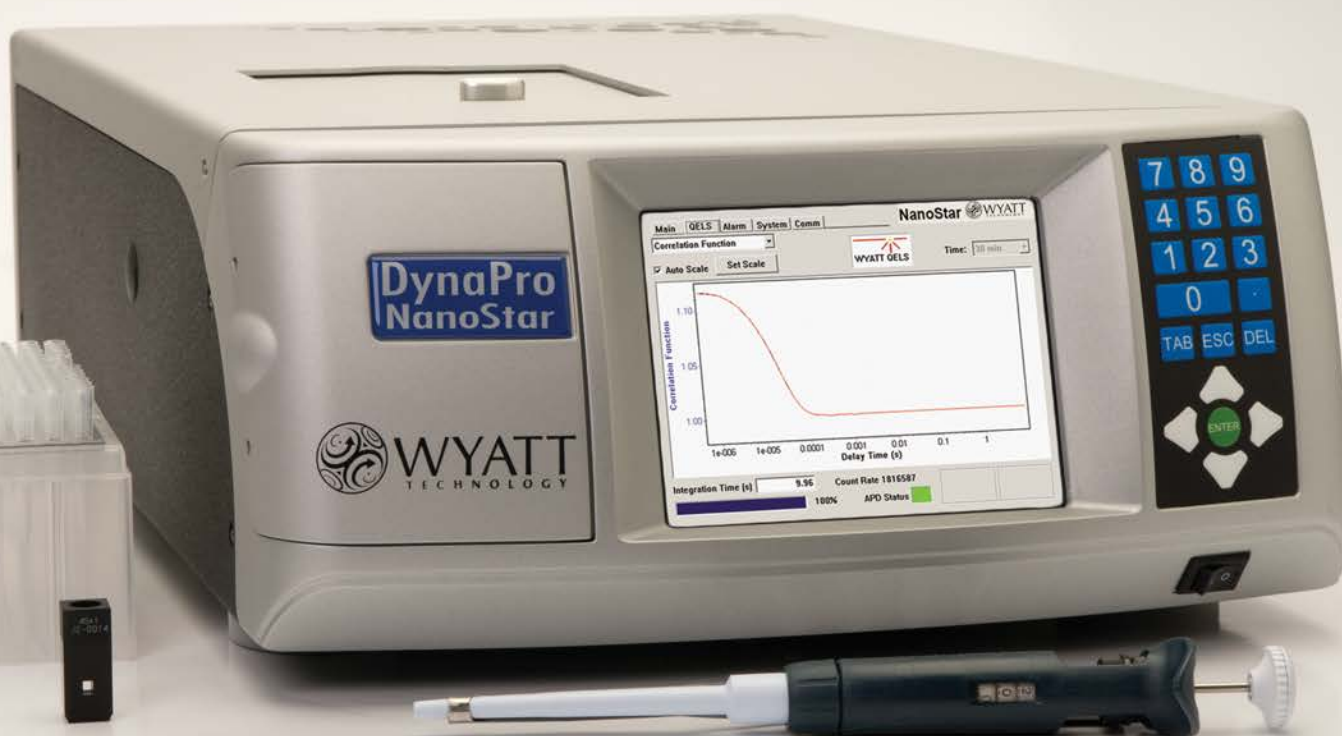




DynaPro® NanoStar®

The most versatile dynamic light scattering instrument for proteins, polymers, and nanoparticles



The DynaPro NanoStar

Excellence in dynamic light scattering

Recognized throughout the world for leading performance in dependability, sensitivity, and small sample volume, the DynaPro dynamic light scattering (DLS) instrument, under the aegis of Wyatt Technology's innovative team of development scientists, takes a giant leap forward in performance with the NanoStar.

Whether achieving quality measurements under the most dilute sample conditions, or over the widest range of controlled thermal conditions, the DynaPro NanoStar meets your highest expectations for the characterization of biomolecules and nanoparticles in solution with DLS.

The NanoStar takes you beyond the performance of conventional batch DLS instruments with its ability to measure *absolute* molecular weight with a dedicated static light scattering detector.

The NanoStar can be attached to the DAWN® HELEOS® or miniDAWN™ TREOS® for simultaneous *on-line* multi-angle light scattering (MALS) and DLS measurements. The NanoStar is used to:

- characterize protein size, hydrodynamic radius, and molecular weight, M_w
- determine size, hydrodynamic radius, and size distributions of nanoparticles
- detect and quantify protein aggregation
- screen protein formulations
- optimize protein crystallization conditions
- study thermal stability
- analyze protein folding/unfolding
- measure size and stability of liposomes
- monitor association equilibrium
- detect compound aggregation
- measure second virial coefficients, A_2

Multi-mode optical fiber advantage

The NanoStar achieves the highest performance in DLS through the perfect alignment of Wyatt's proprietary laser systems, multi-mode optical fiber, and exquisitely sensitive photon detectors.

If you've ever looked at laser light reflected from a surface, you've probably noticed that, unlike other light sources, the intensity pattern from laser light doesn't look smooth. Instead, the pattern shows a granular texture. This is due to the coherent nature of the laser light which, when it arrives at your eye, adds constructively or destructively giving rise to light and dark speckles. When the scattered light comes from a sample within the NanoStar, the speckles fluctuate due to the Brownian motion of the molecules. It is the *time intensity fluctuations* of the speckles that are analyzed by the NanoStar to provide the nanoparticle size and size distributions.

In the NanoStar, a sample is illuminated with a highly stable, coherent laser beam. The light scattered by the sample is collected by an optical fiber where it is detected by an extremely sensitive avalanche photodiode. The question arises, for the best performance, *how many* speckles should the optics collect? There are two types of fibers that are typically used to collect light: single-mode and multi-mode. Single-mode fibers essentially collect light from a single speckle, which gives good contrast in the fluctuation from light to dark, but collects only a very small fraction of the scattered light. Multi-mode fibers, on the other hand, have a slightly reduced contrast, but collect light from several speckles simultaneously. This allows the instrument to gather a much larger fraction of the scattered light. This bigger signal gives better performance for weakly scattering samples, typically the smallest nanoparticles under the most dilute conditions. This is the essence of the multi-mode advantage.

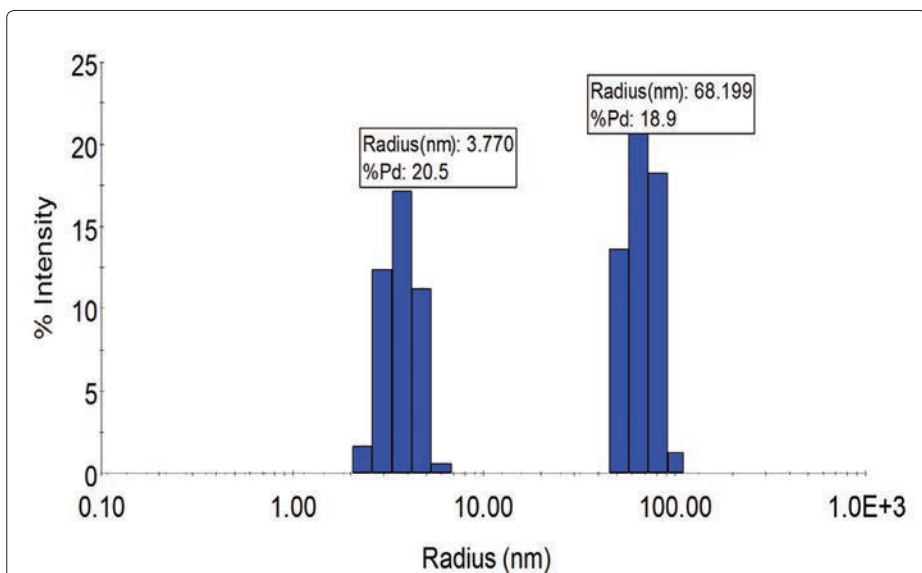
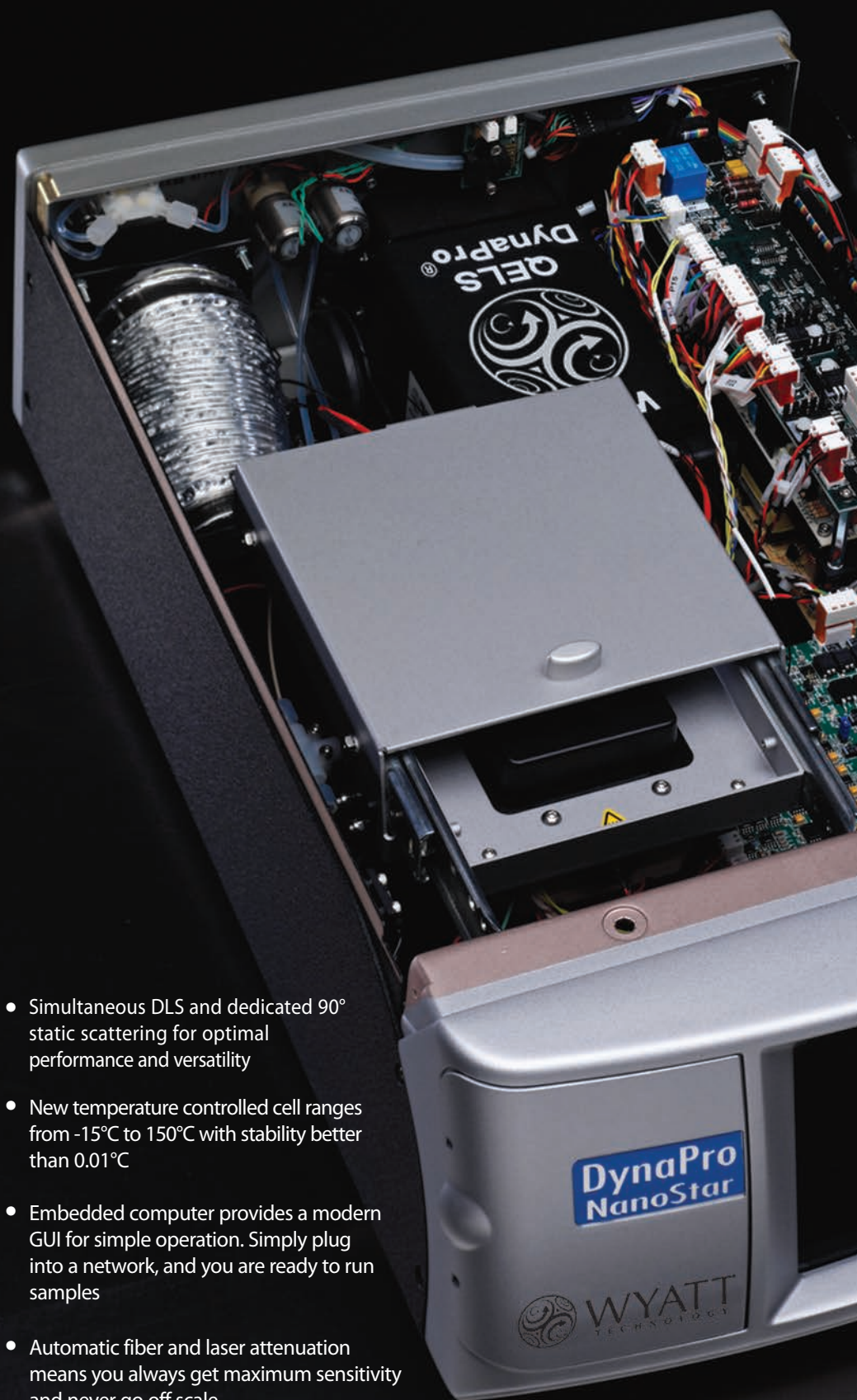


Figure 1. Real time size distributions are displayed in Histogram format with peak label. Customizable tables are generated highlighting the parameters of critical importance to your applications, such as size distribution and polydispersity.



- Simultaneous DLS and dedicated 90° static scattering for optimal performance and versatility
- New temperature controlled cell ranges from -15°C to 150°C with stability better than 0.01°C
- Embedded computer provides a modern GUI for simple operation. Simply plug into a network, and you are ready to run samples
- Automatic fiber and laser attenuation means you always get maximum sensitivity and never go off scale



How It Works...

The light scattered by the sample is collected by a custom optical fiber. The fiber collects wavelets of light, which scatter destructively or constructively, depending on the positions of the illuminated molecules. As the molecules undergo Brownian motion, their relative positions change with time. Small molecules—which diffuse quickly—generate signals that fluctuate **rapidly**. Conversely, large molecules generate signals that fluctuate **slowly**. The time dependence of these fluctuations is characterized by the intensity autocorrelation function which is defined as

$$G(\tau) = \int_{-\infty}^{\infty} I(t)I(t-\tau)dt$$

where $I(t)$ is the detected intensity as a function of time, and τ is a delay time. The autocorrelation function of a monodisperse sample is related to its diffusion constant by

$$G(\tau) = \langle I(t) \rangle^2 (1 + \alpha e^{-2D_t q^2 \tau})$$

where $\langle I(t) \rangle^2$ is the average scattered intensity squared, α is an instrument constant, D_t is the translational diffusion constant, $q = (4\pi n/\lambda_0) \sin(\theta/2)$, n is the index of refraction of the solvent, λ_0 is the wavelength of light in vacuum, and θ is the angle of detection with respect to the incident beam direction.

By analyzing the correlation function, one can measure directly the diffusion constant of the molecule. Furthermore, the Stokes-Einstein relation can be used to define a “hydrodynamic radius”, which is the size of a sphere which diffuses at the same rate as the molecule under study. The Stokes-Einstein relation enables the molecule’s hydrodynamic radius to be determined:

$$R_h = \frac{k_B T}{6\pi\eta D_t}$$

where k_B is Boltzman’s constant, T is the absolute temperature (in Kelvin), and η is the solvent viscosity.

Auto-attenuation advantage

The application of DLS to nanoparticle characterization can also encompass highly scattering samples, typically larger particles or highly concentrated preparations of nanoparticles. When working with highly scattering samples you must be careful not to saturate the avalanche photo diode (APD), which is extremely sensitive, but has limited dynamic range and can be damaged from being exposed to overly-intense scattered light. The NanoStar *expands* the operating range of the detector by automatically continuously attenuating the laser power from 100% to 10% and adjusting an inline optical attenuator to achieve the optimal signal level. The ability of the NanoStar to automatically attenuate the scattering signal by five orders of magnitude ensures the APD never goes off scale.

Dual detectors provide twice the information

By including two light scattering detectors in its optical system, the NanoStar has the unique advantage of measuring hydrodynamic radius and absolute molecular weight *simultaneously*.

When making measurements of dynamic light scattering, the APD is an ideal detector, providing fast response and high sensitivity to fluctuations in scattered light. However, when measuring the molecular weight of a sample, we are interested in the *average* intensity of the scattered light. While the APD works well for measuring fluctuations in scattered light, its response is extremely nonlinear with intensity, and therefore is not the best detector for measuring static light scattering.

The NanoStar includes a complementary detector to the APD to achieve molecular weight measurements. This is an independent silicon PIN photodetector (PIN PD), dedicated to measuring the total (or static) scattering

signal. The PIN PD is ideal for the static light scattering measurements because it is linear over the entire dynamic range of operation, over six orders of magnitude.

By analyzing the time intensity fluctuations with the APD, and total or static light scattering with the PIN PD, the NanoStar provides two critical parameters for characterizing all types of nanoparticles: size and molecular weight. Together, these *dual detectors* provide twice the information as conventional DLS instruments and help you more fully characterize your nanoparticles, all in a single instrument, as shown in Figure 1.

Thermal characterization at warp speed

Now, imagine a revolutionary temperature control capability for the further characterization of your nanoparticles, over the widest range and at the most precise temperatures ever available in batch DLS. The NanoStar's

temperature control system provides a range from -15°C to 150°C, with a temperature stability of better than 0.01°C over the *entire range*. Furthermore, the proprietary thermal design changes temperatures as fast as 15°C/min, stabilizing at a new temperature in minutes with little overshoot, for the most precise thermal melt experiments possible.

Thermal melt experiments can be performed with confidence, because of the NanoStar's intelligent design. Open the NanoStar's articulated air-cooled, temperature controlled sample chamber, and insert the precision cuvette.

The NanoStar's unique oven design ensures the desired temperature is uniform throughout the compartment and cuvette. No need to worry about errors in size introduced by unpredictable temperature gradients. In addition, the sensitive detectors are thermally isolated from the cell chamber, ensuring that only the sample is heated and cooled, for the highest instrument reliability.

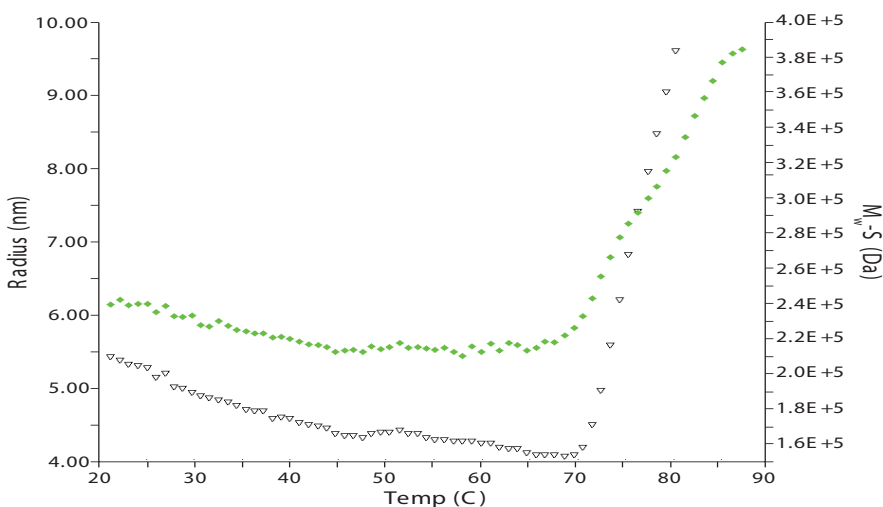


Figure 2. A thermal stability study of an IgG protein using the NanoStar. Radius is determined by dynamic light scattering while molar mass M_w is measured simultaneously with the NanoStar's dedicated static light scattering detector. Prior to the onset of aggregation at 66°C associated with growth of size and M_w , both radius and molar mass decrease with increasing temperature. While size information alone might suggest a conformational change, the additional molecular weight data indicate a thermally-induced dissociation.

When operating *below* ambient temperature, the NanoStar employs a dual valve dry nitrogen flow that keeps the cell dry and dust free. When the sample chamber is opened at low temperatures, a high flow of nitrogen purges and prevents undesirable condensation of moist room air on the optical system. After the cuvette is inserted into the chamber, the NanoStar restricts the flow of nitrogen to conserve gas. A standard lab cylinder will last over a month of continuous usage, but if the cylinder runs out during the middle of an experiment, the NanoStar's internal sensors will detect the lack of nitrogen and automatically raise the temperature, safely avoiding unwanted condensation. Working at room temperature or above? No problem. The valves shut off the nitrogen purge, further conserving gas.

The most advanced custom correlator

The NanoStar employs a custom-designed dedicated correlator. It has a 100 nsec sampling time, which is fast enough to capture all of the dynamics of diffusing particles from less than 0.5 nm, to over 1 μm . Employing a multi-tau architecture with 32 linear bins in the first stage and 16 in subsequent stages at progressively increasing sample times, it has a total of 512 bins that simultaneously measure the autocorrelation function from 100 nsec to 1 hour. This guarantees that polydisperse samples with a wide range of molecular sizes are accurately measured.

Disposable vs. Quartz cuvette data quality

For high quality DLS data at a low cost, we offer a disposable cuvette designed especially for the NanoStar. The disposable cuvettes feature a 4 μL minimum sample volume, with up to 700 μL maximum. The

data quality is comparable to your quartz cuvette, and vastly superior to other disposable cuvettes. The disposable cuvettes have an integrated evaporation control, where the sample chamber is surrounded with a reservoir (moat) which can be filled with solvent to saturate the interior with vapor, thus reducing evaporation from the sample solution. Alternatively, the sample itself can be oil-capped. The reusable silicone cap (plug) can be used to seal the cuvette. The cuvette is constructed of a Cyclic Olefin Copolymer (COC), which is compatible with aqueous buffers and alcohols. The cuvette operates from 0-80°C. When the cell is installed in the NanoStar it activates a microswitch, which allows the instrument to restrict the temperature range, as appropriate. Each cuvette fits directly in the NanoStar cell cavity with no adapter required and is keyed; it can only be inserted in the proper orientation. A molded finger grip allows for easy insertion and removal of cuvette.

Explore new frontiers

Whether or not your research efforts currently make use of dynamic light scattering, the NanoStar's unique capabilities will open new frontiers in the characterization of your novel biomolecules or nanoparticles and their behavior in solution or suspension

Non-perturbing and non-destructive, the NanoStar determines the hydrodynamic radius and absolute molecular weight of your samples *simultaneously*. With this information, the NanoStar provides you with unique insights *beyond* those obtained from conventional DLS technologies. Determine the solution behavior of your samples in terms of size and molecular weight. Proteins, peptides, synthetic polymers, quantum dots, vaccines, surfactant micelles, liposomes, and other nanoparticles can be characterized with minimum sample volumes — quickly, accurately, and completely.

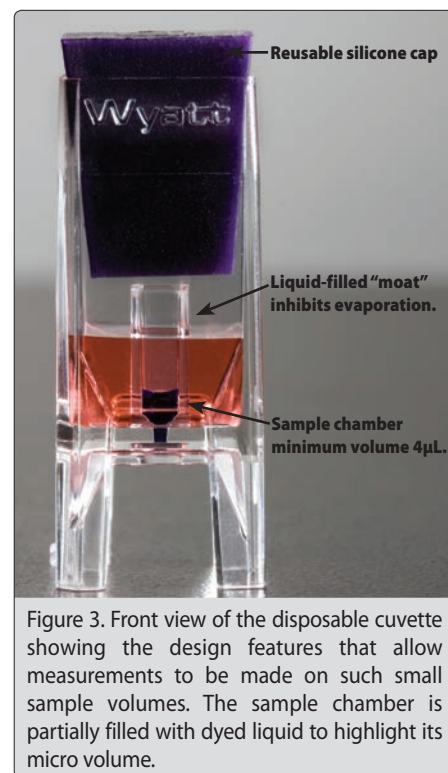


Figure 3. Front view of the disposable cuvette showing the design features that allow measurements to be made on such small sample volumes. The sample chamber is partially filled with dyed liquid to highlight its micro volume.

By providing the highest sensitivity, the NanoStar not only enables the detection of the *smallest* macromolecules but also affords characterization of your samples under novel solution conditions, including dilute concentrations and exotic solvent or buffer conditions. With higher sensitivity, the NanoStar detects trace amounts of aggregation that may go undetected with more conventional DLS instruments. Alternatively, the NanoStar can capture data more quickly than conventional batch technologies, providing more data for kinetics studies of changes in size and molecular weight.

The unusually wide temperature range of the NanoStar offers characterization of your samples at both lower *and* higher temperatures than most DLS instruments, making it ideal for performing rapid thermal melt studies of therapeutic proteins. Other applications include investigating shape-memory polymers or characterizing micelle thermal response.

DYNAMICS® Software

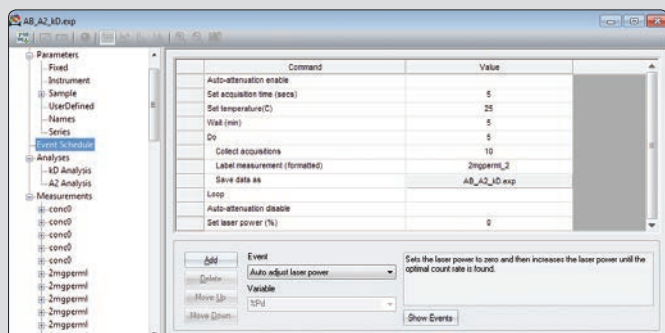


Figure 4. With the DYNAMICS built-in event scheduler, you can easily program the NanoStar to do complex experiments including replicate measurements and melting curve ramps.

Item	Temp (C)	Radius (nm)	Mw-S (kDa)	Conc (mg/mL)	Sample	Acq Time (s)	Span (D90 - D10) (%)
12	40pmml	25.0	5.727	0.0	172.9	4	0.142
13	40pmml	25.0	5.722	-0.0	172.9	4	0.187
14	40pmml	25.0	5.727	0.0	172.9	4	0.181
15	40pmml	25.0	5.704	0.0	172.9	4	0.137
16	40pmml	25.0	5.875	0.0	175.4	6	0.255
17	40pmml	25.0	5.881	-0.0	175.7	6	0.267
18	40pmml	25.0	5.881	-0.0	175.7	6	0.278
19	40pmml	25.0	5.885	0.0	175.7	6	0.258
20	40pmml	25.0	5.873	-0.0	175.6	6	0.147
21	40pmml	25.0	6.061	0.0	183.2	8	0.218
22	40pmml	25.0	6.011	0.1	183.2	8	0.326
23	40pmml	25.0	6.010	-0.0	183.2	8	0.417
24	40pmml	25.0	6.056	0.0	183.3	8	0.328
25	40pmml	25.0	6.052	0.0	183.7	8	0.371
26	10pmml	25.0	6.184	-0.0	188.5	10	0.331
27	10pmml	25.0	6.240	0.0	188.6	10	0.336
28	10pmml	25.0	6.256	0.0	188.6	10	0.337
29	10pmml	25.0	6.212	-0.0	188.6	10	0.275
30	10pmml	25.0	6.239	0.0	188.6	10	0.291
31	12mpmml	25.0	6.321	-0.0	188.6	12	0.331
32	12mpmml	25.0	6.306	0.0	188.7	12	0.267
Mean	---	25.0	6.261	0.3	187.6	10	0.276
Std	---	0.0	0.442	0.6	11.3	6	0.1
%S	---	0.0	7.000	236.3	6.0	66	36.4
Min	---	25.0	5.538	-0.0	169.1	0	0.0766
Max	---	25.0	6.962	2.8	209.1	20	0.531

Figure 5. View real time results in tabular format, including absolute M_w from the static LS detector and R_h from the dynamic LS detector inside the NanoStar. Statistics of the results are automatically calculated.

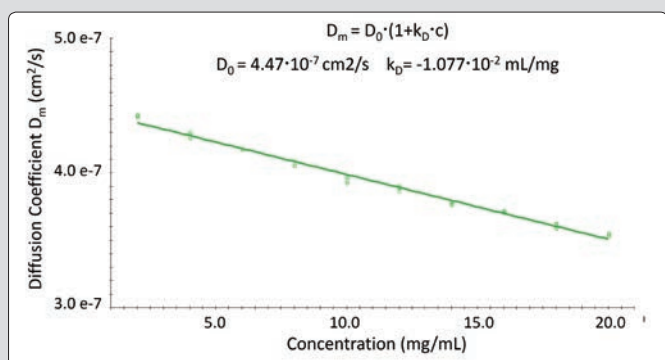


Figure 6. Colloidal stability of proteins across a range of formulation conditions is captured in the concentration dependence of the diffusion coefficient, known as k_D . A k_D analysis is shown here of a monoclonal antibody. In the same experiment, the Static LS detector allows for the determination of the second virial coefficient A_2 , as shown in Figure 7.

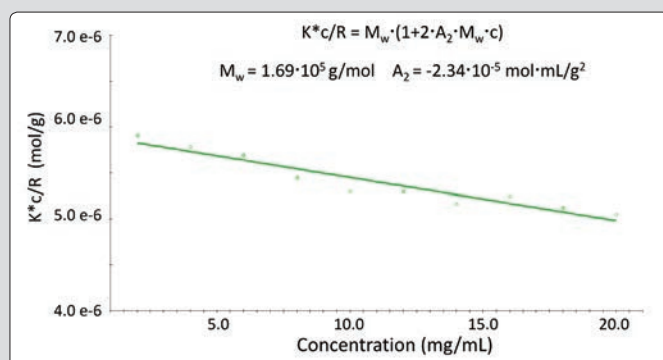


Figure 7. DYNAMICS software allows you to quantify protein molecular interactions by determining the second virial coefficient A_2 from the change in Static LS intensity as a function of concentration. A_2 can provide useful indication of formulation stability. An A_2 analysis is shown here of a monoclonal antibody that experiences a net attraction.

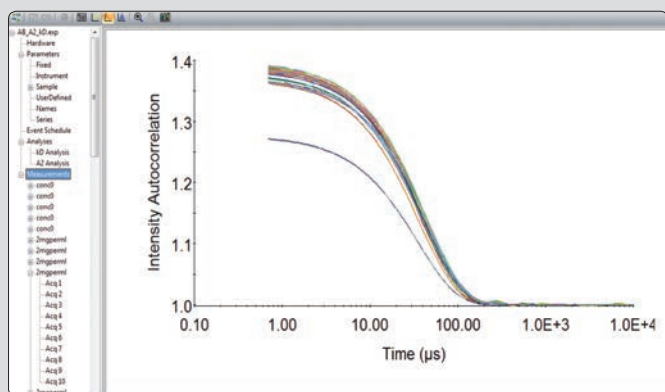


Figure 8. DYNAMICS software is designed to manage and analyze large amounts of data, including the unique ability to overlay correlation functions and size distributions from any or all of the data collected.

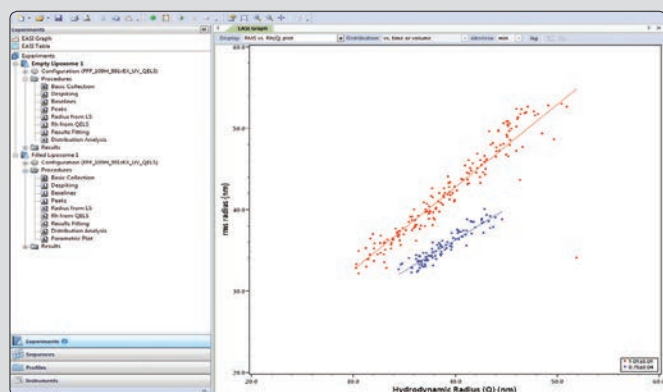


Figure 9. The DynaPro NanoStar was used with a HELEOS MALS detector to simultaneously measure root-mean square (RMS) radius and hydrodynamic radius (R_h) of two liposome samples following a field-flow fractionation (FFF) separation. RMS radius was plotted against R_h for empty liposomes (red) and filled liposomes (blue). The ratio is close to 1 for empty liposomes and 0.77 for filled liposomes, consistent with the theoretical values for these two structures.

Additional DLS Applications & Products

DynaPro Plate Reader

The DynaPro Plate Reader II performs fully automated analysis of nanoparticle size, aggregation and stability, non-invasively, by dynamic light scattering. Measurements take place in situ in industry-standard, disposable, microtiter plates of 96, 384, or 1536 wells. Thermal stability can be studied on hundreds of samples and buffer conditions simultaneously over 4 - 85°C. An embedded camera photographs each well to assess precipitation or identify bubbles.



The DynaPro Plate Reader automates DLS measurements for the highest levels of productivity and reliability.

Möbiuž®

The Möbiuž is the *first* and only electrophoretic light scattering (ELS) instrument for reliable, reproducible, and non-destructive electrophoretic mobility measurements of nanoparticles and macromolecules as small as 2 nm, under diverse solution conditions. The Möbiuž measures hydrodynamic radius via DLS, simultaneously and in parallel with mobility, to determine zeta potential or molecular charge.

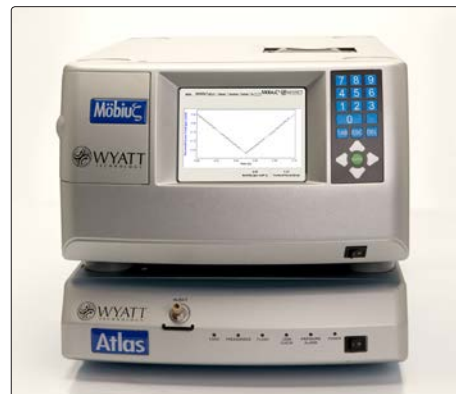
Möbiuž with an Autosampler

The flow cell of the Möbiuž is uniquely designed as a closed system (without free fluid surfaces) to accept sample delivery from an autosampler. Once the sample injection sequence is implemented and synchronized

with the data acquisition schedule in DYNAMICS, automated multi-sample measurements of electrophoretic mobility become a reality. Clean buffer injections between samples minimize possible cross contamination. After the measurements are complete, a wash sequence can be added to ensure a clean cell for the next round of measurements.

Atlas™ Cell Pressurization System

Electrical measurements in conducting solutions are often disrupted by gas bubbles generated via electrolysis. Wyatt Technology's Atlas system pressurizes the Möbiuž cell to 35 bar (500 psi). This forces evolved gas back



The Möbiuž and Atlas cell pressurization system can be integrated with an autosampler for enhanced productivity.

into solution to eliminate bubble formation, for mobility analysis at ionic strengths up to 8x that of physiological saline.

PLATE READER BENEFITS

Experiment	Description	Batch (Cuvette) Operation	Plate Reader (Microplate)
Measurement of 1536 distinct samples	Loading samples	> 48 hrs (includes time to insert, remove, and clean each cuvette)	< 2 hrs (1536 well plate)
	Collecting data, personnel time required	> 24 hrs (personnel continuously required to change samples)	No personnel required
	Collecting data, measurement time	13 hrs	13 hrs
	Total personnel time required	> 72 hrs	< 2 hrs
Measurement of temperature dependence (42 different temperatures) of 384 samples	Loading samples	> 36 hrs (includes time to insert, remove, and clean each cuvette, and wait for temperature stability)	< 1 hr
	Collecting data, measurement time	634 hrs (> 26 days), with a person required to change samples every two hrs	45 hrs (no personnel required)
	Totals (personnel time only)	28 days (36 hrs)	3.25 days (< 1 hr)

With the DynaPro Plate Reader, a temperature-dependent experiment using a 384-well plate requires < 1 hr to fill the wells and 45 hrs (unattended) to acquire the data. The same data acquired with a single cuvette-based instrument would require personnel to change the sample every 2 hrs over > 28 days total run time.

Specifications

	DynaPro NanoStar	Plate Reader II
Detection Channels	Dynamic Light Scattering Static Light Scattering	Dynamic Light Scattering Photograph of well
Performance		
Size Range R_h by DLS	0.25 - 2500 nm	0.5 - 1000 nm
Molar Mass Range by SLS	300 Da to 1 MDa (concentration dependent)	N/A
Sensitivity (lysozyme 14.4 kDa)	0.1 mg/mL	0.125 mg/mL
Temperature Range	-15°C to +150°C †	4°C to 85°C †
Sample Options		
Number of Samples	1	1 to 96, 384, or 1536
Minimum Sample	1.25 µL in quartz cuvette	4 µL*
Sample Cell Volume	Quartz cuvette: 1 µL or 45 µL Disposable cuvette: 4 µL, COC (Cyclic Olefin Copolymer)	Industry standard plates 96, 384 or 1536 microwell plates COC, polystyrene, glass
Optics		
Laser Wavelength	658 or 785 nm	830 nm
Laser Power Control	Programmable 10%-100%	Programmable 10%-100%
Scattering Angle	90°	158°
Optical Fiber	Multi-Mode	Multi-Mode
Electronics		
Data Acquisition Time	1 to 3600 sec	1 to 3600 sec; 5-20 sec per well typical
Correlator	512 channel, 100 nsec sampling time in a multi-tau layout	512 channel, 100 nsec sampling time in a multi-tau layout
Dimensions	60 cm (l) x 36 cm (w) x 18 cm (h)	60 cm (l) x 36 cm (w) x 25 cm (h)

* Minimum volume specified with 2 mg/mL Lysozyme in a 1536 well plate, Wyatt P/N P8803-1536, with 25 seconds acquisition.

† Dry air or nitrogen required for operation below 20°C.

Wyatt Technology is committed to continual improvement. Specifications subject to change without notice.

WARRANTY: All Wyatt instruments are guaranteed against manufacturing defects for 1 year.



With installations in *more* than 65 countries, *more* than 11,000 refereed journal publications citing its instruments, and 18+ PhD scientists on site, Wyatt Technology is the **world's leading manufacturer of instruments** for absolute macromolecular characterization. It is the only company in the world focused exclusively on such systems, their design, and their applications.

DAWN, HELEOS, TREOS, Optilab, ViscoStar, NanoStar, Calypso, Möbius, MöbiuZ, ASTRA, DynaPro, DYNAMICS, Aurora, International Light Scattering Colloquium, Light Scattering University, Light Scattering for the Masses, Protein Solutions, Wyatt Technology are registered trademarks of Wyatt Technology Corporation. Wyatt Technology instruments, components and software are covered by one or more of the following: U.S. Patent Nos.: 6,411,383; 6,426,794; 6,452,672; 6,519,032; 6,651,009; 6,774,994; 6,819,420; 6,975,392; 7,027,138; 7,283,221; 7,331,218; 7,386,427; 7,813,882; 7,911,594; 7,982,875; 8,195,405; 8,441,638; 8,525,991; British Patent Nos.: EP 0 710 831; EP 0 665 433; EP 1 134 577; EP 1 510 807; EP 1 517 143; EP 1 538 435; EP 1 507 136; EP 1 645 864; French Patent Nos.: EP 1 517 143; EP 1 645 864; German Patent Nos.: 694 30 918.4-08; 694 33 615.7-08; 601 31 486.7-08; 603 19 078.2-08; 60 2004 022 625.4-08; 60 2004 038 882.3; 60 2004 039 666.4; 60 2005 040 312.4; Japanese Patent Nos.: 4,439,211; 4,381,914; 4,426,951; 4,594,206; 4,680,402; 4,786,906; 4,813,784; 5,261,720; 5,500,365; Chinese Patent Nos.: ZL 2004 1 0070894.8; ZL 2004 1 0080545.4; ZL 2004 1 0070023.6; ZL 2004 1 0062673.6; ZL 2005 1 0108269.2; ZL 2011 1 0008100.5; Korean Patent No.: 794,478. Other patents pending. No part of this brochure may be reproduced in any way without written permission from Wyatt Technology Corporation.

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